

Lecture 10

September 24

Bayes...

Quiz 1: October 8

- Do you need to be able to derive complicated trigonometric formulas?
[REDACTED]
- Do you need to remember exactly which of Mayer's equations were in each group?
[REDACTED]
- Do you need to be able to talk about what Mayer did vs what Boscovich did?
[REDACTED]
- Do you need to know what a probability is?
[REDACTED]
- Do you need to know how to manipulate joint and conditional probabilities?
[REDACTED]
- Do you need to understand what "insufficient reason" means?
[REDACTED]
- Do you need to know biographies?
[REDACTED]
- Do you need to know something about someone?
[REDACTED]

Learning objectives

- Use the two-bag example to compare a rule's performance across repeated samples with the posterior probability that its answer is correct after observing a draw, and explain why choosing a rule requires an evaluation criterion.
- Describe Bayes' table experiment and explain how the position of the first ball determines an unknown success probability while subsequent balls produce a binomial count.
- Use Bayes' fair-value argument to derive the relationship between joint and conditional probabilities, identifying where the probability of the first event enters the calculation.
- Explain how a prior distribution and a model for the data given the parameter combine to produce a posterior distribution

Metaphysical difference of opinion

Frequentist / Legendre
 $p(\text{data} | \text{parameter})$

Should one choose to
evaluate a statistical
procedure before the
observations are made.

Understanding based on
repeated samples

Bayesian / Laplace
 $p(\text{parameter} | \text{data})$

Should one make this
choice a posteriori?
Conditional on the
Observation

What's the difference?

Two bags Mostly blue (8 blue, 2 red)
 Mostly red (2 blue, 8 red)

Rule A: Follow the ball (if you see blue, guess blue)
 if you see red, guess red

Rule B: Always guess blue

Frequentist evaluation

What bag is hidden

Mostly blue

Mostly red

Rule A

80%

88%

Long run prob.

Repeated sampling prob.

Rule B

100%

0%

Which one is better?

Bayesian evaluation

Say you believe there's a 50/50 chance of each bag.

If you see a blue ball and you use Rule A

$$P(\text{mostly blue} \mid \text{a blue ball}) = 0.8$$

$$P(\text{mostly red} \mid \text{a red ball}) = 0.8$$

What about Rule B?

wrong 50% of the time

Bayes and Bernoulli's problem

PROBLEM

Given the number of times in which an unknown event has happened and failed: Required the chance that the probability of its happening in a single trial lies somewhere between any two degrees of probability that can be named.

Given $X \sim \text{Binom}(n, p)$ $\text{Binom}(n, \theta)$

Required $P(a \leq \theta \leq b \mid X=x)$ can be calculated
 $\uparrow$
less than

Bernoulli $\left| \frac{X}{n} - \frac{S}{S+T} \right| < \epsilon$
 S is # of blue balls
 T is # of red balls

THE URN ★ ★ ★ ★ ★ VS. ★ ★ ★ ★ ★ THE TABLE

DIFFERENT FORMS. A HIGHER PURPOSE.



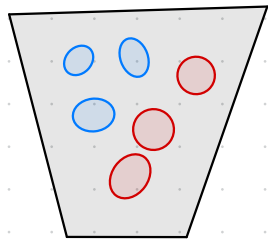
THINGS • IDEAS • PEOPLE • PERSPECTIVE



The urn

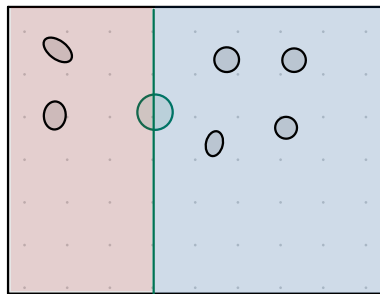
vs

The table



Pull one ball
at a time, record
the outcome, return
to the urn,
repeat

$$X \sim \text{Binom}\left(n, \frac{b}{r+b}\right)$$



Roll  green ball
Then roll other balls
and count how many are
in  vs 
sections of the table

$$X \sim \text{Binom}\left(n, \frac{\text{blue square}}{\text{red square} + \text{blue square}}\right)$$

Same model if $\frac{b}{r+b} = \frac{\text{blue square}}{\text{red square} + \text{blue square}}$

The table

The first ball is uniform. The area to the right of it is θ

The next n balls are uniform, and probability of being to the right of first ball is θ .

Data generating process

Prior

— $p(\theta)$ my prior beliefs about a parameter

Model

— $p(X|\theta)$ how my data are generated

Posterior

— $p(\theta|X)$ the update to my beliefs after seeing the data

Why the uniform?

Is θ really uniform due to "insufficient reason"?

$$p(\theta) = 1 \quad \text{when } \theta \in [0,1] \quad \text{w/ 0 otherwise}$$

$\uparrow$
in

$$P(X=x) = \frac{1}{n+1}$$

$\uparrow$
no conditionally

marginal probability of a Success

[If I don't know θ then all outcomes of X are equally likely.

Bayes fundamental view on probability, an aside

$$P(E_2 | E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)}, \text{ why?}$$

Next class